

NAME: _____ PERIOD: _____ DATE: _____

Numbers That Aren't Quantities

AP Statistics · Topic 1.2 (CED 1.2) · B: 2026-09-10 · E: 2026-09-11

[STAR] TODAY'S PROBLEM

A school newspaper surveyed six students for a story about getting to school. Part of the data table is shown.

Two reporters disagree about the numbers in it. **Maya:** “Zip code and Rating are both quantitative — they’re numbers, so we can average them.” **Jordan:** “They’re both categorical. Numbers can be labels.”

Getting-to-school survey (six students)

Student	Lunch period	Zip code	Commute (min)	Rating (1–5 stars)
Aiden	A	01902	12	4
Bea	B	01905	35	2
Cruz	A	01902	8	5
Dev	C	01904	22	3
Elise	B	01905	41	1
Fiona	A	01901	15	4

FIRST TAKE — before the video (5 min)

Is a zip code a quantitative variable? One sentence: yes or no, and the reason you’d give Maya.

[BOOK] VARIABLES (keep on the page)

An **individual** is one row: the person or thing the data describe. A **variable** is one column: a characteristic that changes from individual to individual.

Categorical — the values are category names or group labels (even if written with digits).

Quantitative — the values are numbers for a *measured or counted quantity*, with a unit. Test: does “twice as big” or “the average” mean anything? If not, it’s a label.

Finish after the video:

DOK 1 (a) Who are the individuals in this data set, and what are the four variables?

DOK 2 (b) Classify each of the four variables as categorical or quantitative. For each quantitative variable, state its unit.

DOK 3 (c) Decide who is right about **Zip code** and who is right about **Rating**, using the definition of a quantitative variable in each case. Then name one specific calculation a reader would find meaningless if Maya's classification were used for Zip code, and explain why it is meaningless.

Frame: Zip code is _____ because a quantitative variable must be a _____, and a zip code is a _____.

Frame: A _____ of zip codes would be meaningless because _____.

EXIT — turn this in

One thing the video changed about my first take: _____

Optional #1 — not collected

DOK 2 For each variable, write **C** (categorical) or **Q** (quantitative); for each Q, give the unit: (1) jersey number of a soccer player; (2) number of goals scored this season; (3) shoe size; (4) preferred position (goalkeeper, defender, midfielder, forward); (5) minutes played in the last match.